

VIT BHOPAL UNIVERSITY



LIBRARY MANAGEMENT SYSTEM

NAME: Tanisha Gagan Bihari Parida

REGISTRATION NO.: 26BAI10871

FACULTY NAME: Ashfaq Ahmad Najar

COURSE: BTech CSE (Specialization AI/ML)

SCHOOL: SCAI

Library Management System

1. Introduction

Library Management System is a simple Python based project used to store and manage books in a library. The project is console based and uses basic Python concepts such as variables, functions, lists, dictionaries, loops and conditional statements.

2. Problem Statement

Managing books manually can take time and may lead to mistakes while keeping records. It can also become difficult to keep track of available and issued books when there are multiple books.

This project provides a simple way to add books, display book records, search for books, issue books and return books.

3. Objectives

- Store book details.
- Display all book records.
- Search for a book.
- Issue a book.
- Return a book.
- Delete a book.
- Manage book availability.

4. Functional Requirements

1. **Book Management:** Add, display, search and delete book records.
2. **Book Issue:** Issue an available book and change its status.
3. **Book Return:** Return an issued book and change its status back to available.
4. **Book Search:** Search for a book using its book ID.

5. Non-Functional Requirements

- **Usability:** Simple menu and clear output.

- **Reliability:** Correctly manages book records and book status.
- **Error Handling:** Handles invalid choices and unavailable book records.
- **Maintainability:** Functions keep different tasks separate and easy to modify.

6. System Architecture

USER → MAIN MENU / PROGRAM → BOOK MANAGEMENT / BOOK SEARCH / BOOK ISSUE & RETURN → BOOK RECORDS

7. Workflow / Process

START → Display Menu → Enter Choice → Perform Selected Operation → Display Result → Return to Menu → Exit

8. Use Case

The user can add books, display books, search for a book, issue a book, return a book, delete a book and exit the system.

9. Sequence

User → Select operation → System asks for input → System processes book data → System displays result.

10. Component Design

- Main Menu
- Book Management
- Book Search
- Book Issue and Return
- Book Records

11. Design Decisions

Basic Python concepts were selected to demonstrate fundamental programming skills. Functions divide the program into smaller tasks. A dictionary stores book records and lists store book details such as title, author and status.

A menu-driven approach makes the system simple and easy to use.

12. Implementation

The project is implemented in Python. A while loop keeps the menu running. Functions are used for adding, displaying, searching and deleting books, as well as issuing and returning books.

If-else statements are used to check book availability and for loops are used to process book records.

13. Sample Results

Example:

Book ID: 101

Title: Python Basics

Author: ABC

Status: Available

The program can also search for the book, issue the book, return the book and delete the book.

14. Testing

Test cases include adding a book, displaying books, searching for an existing book, searching for an unknown book, issuing an available book, issuing an already issued book, returning a book, deleting a book, invalid menu choice and exiting.

15. Challenges Faced

- Understanding how to store multiple book records.
- Managing book availability.
- Understanding how to issue and return books.
- Creating a menu using a while loop.
- Dividing the program into separate functions.
- Managing book records using a dictionary.

16. Learnings

Learned functions, lists, dictionaries, loops, conditions, input and output, menu-driven programming, basic testing and project documentation.

17. Future Enhancements

- Add student or member details.
- Store records in a file.
- Add book issue dates.
- Add return dates.
- Add more search options.
- Add a graphical user interface.
- Add a database for storing records.

18. Conclusion

The Library Management System provides a simple way to manage basic library operations using Python programming concepts.

It demonstrates functions, lists, dictionaries, loops, conditions and menu-driven programming in a practical project.

19. References

1. Python documentation.
2. CSE course notes.
3. VITyarthi Build Your Own Project guidelines.